

Elbow Method + Silhouette Method Combination

✅ How to Combine Them

Step 1: Use the Elbow Method to Narrow Down k

- Run K-Means for a range of clusters (say $k = 2$ to 10).
 - Plot SSE (inertia) vs. k .
 - Look for the **elbow point** where improvement slows.
 - Usually you'll see **2–3 candidate k values** near the elbow.
Example: The elbow looks around $k = 3$ or 4 .
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Step 2: Use Silhouette Method on Candidate k Values

- Now only check silhouette scores for those candidate values (instead of all possible k).
 - The silhouette score gives you a **numerical, objective measure** of how well-separated the clusters are.
 - Choose the k with the **highest silhouette score**.
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⚡ Example

Suppose we tested $k = 2$ to 10 .

- **Elbow Method Plot** shows the elbow around $k = 3$ or 4 .
- **Silhouette Scores:**
 - $k = 3 \rightarrow 0.57$
 - $k = 4 \rightarrow 0.64$

So we pick $k = 4$ (best silhouette score among elbow candidates).

How This Works in Practice

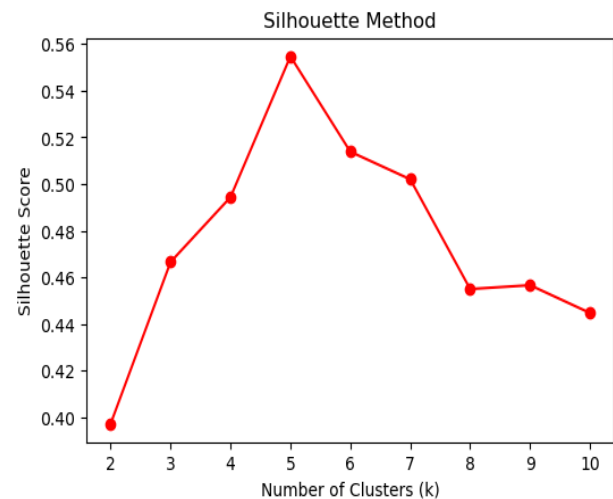
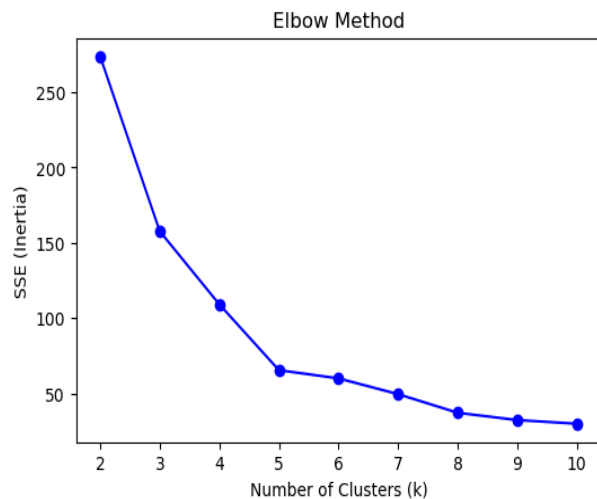
1. Elbow Method:

- You'll see SSE dropping sharply until around **k=5**, then the improvement slows → suggests **5 clusters**.

2. Silhouette Method:

- Prints values like:
 - $k=4 \rightarrow 0.55$
 - $k=5 \rightarrow 0.62$ ✓ (highest)
 - $k=6 \rightarrow 0.48$
- So **k=5** is chosen.

✓ **Result:** You discover there are **5 customer segments** (e.g., low income–low spenders, high income–high spenders, etc.).



Why Use Both?

- **Elbow alone** → subjective, not always clear where the bend is.
- **Silhouette alone** → computationally heavier for large datasets.
- **Together** → elbow gives a small candidate range, silhouette gives the final objective choice.